u_sw

switch_filter

clk

sw_in[12:0]

sw_out[12:0]

cnt[0]_i

RTL_MUX S

I0[3:0] S=1'b1

I1[3:0] S=default O[3:0]

cnt[0]_i__0

RTL_ROM

O A[4:0]

cnt[0]_i__1

RTL_ROM

O A[3:0]

cnt[0]_i__3

RTL_MUX

I0 S=16'b1100001101001111
I1 S=default

O

S[15:0]

cnt[0]0_i +RTL_ADD

I1

I0[3:0] O[3:0] -

cnt[0]0_i__0

RTL_SUB

I0[3:0]

I1[3:0] V=B"0001" O[3:0]

cnt[1]_i

RTL_MUX S

I0[3:0] S=1'b1

I1[3:0] S=default O[3:0]

cnt[1]_i__0

RTL_ROM

O A[4:0]

cnt[1]_i__1

RTL_ROM

O A[3:0]

cnt[1]_i__3

RTL_MUX

I0 S=16'b1100001101001111
I1 S=default

O

S[15:0]

cnt[1]0_i +RTL_ADD

I1

I0[3:0] O[3:0] -

cnt[1]0_i__0

RTL_SUB

I0[3:0]

I1[3:0] V=B"0001" O[3:0]

cnt[2]_i

RTL_MUX S

I0[3:0] S=1'b1

I1[3:0] S=default O[3:0]

cnt[2]_i__0

RTL_ROM

O A[4:0]

cnt[2]_i__1

RTL_ROM

O A[3:0]

cnt[2]_i__3

RTL_MUX

I0 S=16'b1100001101001111
I1 S=default

O

S[15:0] cnt[2]0_i +RTL_ADD

I1

I0[3:0] O[3:0] -

cnt[2]0_i__0

RTL_SUB

I0[3:0]

I1[3:0] V=B"0001" O[3:0]

cnt[3]_i

RTL_MUX S

I0[3:0] S=1'b1

I1[3:0] S=default O[3:0]

cnt[3]_i__0

RTL_ROM

O A[4:0]

cnt[3]_i__1

RTL_ROM

O A[3:0]

cnt[3]_i__3

RTL_MUX

I0 S=16'b1100001101001111
I1 S=default

O

S[15:0]

cnt[3]0_i +RTL_ADD

I1

I0[3:0] O[3:0] -

cnt[3]0_i__0

RTL_SUB

I0[3:0]

I1[3:0] V=B"0001" O[3:0]

cnt[4]_i

RTL_MUX S

I0[3:0] S=1'b1

I1[3:0] S=default O[3:0]

cnt[4]_i__0

RTL_ROM

O A[4:0]

cnt[4]_i__1

RTL_ROM

O A[3:0]

cnt[4]_i__3

RTL_MUX

I0 S=16'b1100001101001111
I1 S=default

O

S[15:0] cnt[4]0_i +RTL_ADD

I1

I0[3:0] O[3:0] -

cnt[4]0_i__0

RTL_SUB

I0[3:0]

I1[3:0] V=B"0001" O[3:0]

cnt[5]_i

RTL_MUX S

I0[3:0] S=1'b1

I1[3:0] S=default O[3:0]

cnt[5]_i__0

RTL_ROM

O A[4:0]

cnt[5]_i__1

RTL_ROM

O A[3:0]

cnt[5]_i__3

RTL_MUX

I0 S=16'b1100001101001111
I1 S=default

O

S[15:0]

cnt[5]0_i +RTL_ADD

I1

I0[3:0] O[3:0] -

cnt[5]0_i__0

RTL_SUB

I0[3:0]

I1[3:0] V=B"0001" O[3:0]

cnt[6]_i

RTL_MUX S

I0[3:0] S=1'b1

I1[3:0] S=default O[3:0]

cnt[6]_i__0

RTL_ROM

O A[4:0]

cnt[6]_i__1

RTL_ROM

O A[3:0]

cnt[6]_i__3

RTL_MUX

I0 S=16'b1100001101001111
I1 S=default

O

S[15:0]

cnt[6]0_i +RTL_ADD

I1

I0[3:0] O[3:0] -

cnt[6]0_i__0

RTL_SUB

I0[3:0]

I1[3:0] V=B"0001" O[3:0]

cnt[7]_i

RTL_MUX S

I0[3:0] S=1'b1

I1[3:0] S=default O[3:0]

cnt[7]_i__0

RTL_ROM

O A[4:0]

cnt[7]_i__1

RTL_ROM

O A[3:0]

cnt[7]_i__3

RTL_MUX

I0 S=16'b1100001101001111
I1 S=default

O

S[15:0]

cnt[7]0_i +RTL_ADD

I1

I0[3:0] O[3:0] -

cnt[7]0_i__0

RTL_SUB

I0[3:0]

I1[3:0] V=B"0001" O[3:0]

cnt[8]_i

RTL_MUX S

I0[3:0] S=1'b1

I1[3:0] S=default O[3:0]

cnt[8]_i__0

RTL_ROM

O A[4:0]

cnt[8]_i__1

RTL_ROM

O A[3:0]

cnt[8]_i__3

RTL_MUX

I0 S=16'b1100001101001111
I1 S=default

O

S[15:0]

cnt[8]0_i +RTL_ADD

I1

I0[3:0] O[3:0] -

cnt[8]0_i__0

RTL_SUB

I0[3:0]

I1[3:0] V=B"0001" O[3:0]

cnt[9]_i

RTL_MUX S

I0[3:0] S=1'b1

I1[3:0] S=default O[3:0]

cnt[9]_i__0

RTL_ROM

O A[4:0]

cnt[9]_i__1

RTL_ROM

O A[3:0]

cnt[9]_i__3

RTL_MUX

I0 S=16'b1100001101001111
I1 S=default

O

S[15:0]

cnt[9]0_i +RTL_ADD

I1

I0[3:0] O[3:0] -

cnt[9]0_i__0

RTL_SUB

I0[3:0]

I1[3:0] V=B"0001" O[3:0]

cnt[10]_i

RTL_MUX S

I0[3:0] S=1'b1

I1[3:0] S=default O[3:0]

cnt[10]_i__0

RTL_ROM

O A[4:0]

cnt[10]_i__1

RTL_ROM

O A[3:0]

cnt[10]_i__3

RTL_MUX

I0 S=16'b1100001101001111
I1 S=default

O

S[15:0]

cnt[10]0_i +RTL_ADD

I1

I0[3:0] O[3:0] -

cnt[10]0_i__0

RTL_SUB

I0[3:0]

I1[3:0] V=B"0001" O[3:0]

cnt[11]_i

RTL_MUX S

I0[3:0] S=1'b1

I1[3:0] S=default O[3:0]

cnt[11]_i__0

RTL_ROM

O A[4:0]

cnt[11]_i__1

RTL_ROM

O A[3:0]

cnt[11]_i__3

RTL_MUX

I0 S=16'b1100001101001111
I1 S=default

O

S[15:0]

cnt[11]0_i +RTL_ADD

I1

I0[3:0] O[3:0] -

cnt[11]0_i__0

RTL_SUB

I0[3:0]

I1[3:0] V=B"0001" O[3:0]

cnt[12:0]_i__2

RTL_MUX

I0 S=1'b1
I1 S=default

O S

cnt[12]_i

RTL_MUX S

I0[3:0] S=1'b1

I1[3:0] S=default O[3:0]

cnt[12]_i__0

RTL_ROM

O A[4:0]

cnt[12]_i__1

RTL_ROM

O A[3:0]

cnt[12]_i__3

RTL_MUX

I0 S=16'b1100001101001111
I1 S=default

O

S[15:0]

cnt[12]0_i +RTL_ADD

I1

I0[3:0] O[3:0] -

cnt[12]0_i__0

RTL_SUB

I0[3:0]

I1[3:0] V=B"0001" O[3:0]

cnt_reg[0][3:0]

RTL_REG C

CED Q

cnt_reg[1][3:0]

RTL_REG C

CED Q

cnt_reg[2][3:0]

RTL_REG C

CED Q

cnt_reg[3][3:0]

RTL_REG C

CED Q

cnt_reg[4][3:0]

RTL_REG C

CED Q

cnt_reg[5][3:0]

RTL_REG C

CED Q

cnt_reg[6][3:0]

RTL_REG C

CED Q

cnt_reg[7][3:0]

RTL_REG C

CED Q

cnt_reg[8][3:0]

RTL_REG C

CED Q

cnt_reg[9][3:0]

RTL_REG C

CED Q

cnt_reg[10][3:0]

RTL_REG C

CED Q

cnt_reg[11][3:0]

RTL_REG C

CED Q

cnt_reg[12][3:0]

RTL_REG C

CED Q

q_i

RTL_ROM

O A[4:0]

q_i__0

RTL_MUX

I0 S=1'b1
I1 S=default

O S
q_i__1

RTL_ROM

O A[4:0]

q_i__2

RTL_MUX

I0 S=1'b1
I1 S=default

O S
q_i__3

RTL_ROM

O A[4:0]

q_i__4

RTL_MUX

I0 S=1'b1
I1 S=default

O S
q_i__5

RTL_ROM

O A[4:0]

q_i__6

RTL_MUX

I0 S=1'b1
I1 S=default

O S

q_i__7

RTL_ROM

O A[4:0]

q_i__8

RTL_MUX

I0 S=1'b1
I1 S=default

O S

q_i__9

RTL_ROM

O A[4:0]

q_i__10

RTL_MUX

I0 S=1'b1
I1 S=default

O S
q_i__11

RTL_ROM

O A[4:0]

q_i__12

RTL_MUX

I0 S=1'b1
I1 S=default

O S
q_i__13

RTL_ROM

O A[4:0]

q_i__14

RTL_MUX

I0 S=1'b1
I1 S=default

O S

q_i__15

RTL_ROM

O A[4:0]

q_i__16

RTL_MUX

I0 S=1'b1
I1 S=default

O S

q_i__17

RTL_ROM

O A[4:0]

q_i__18

RTL_MUX

I0 S=1'b1
I1 S=default

O S

q_i__19

RTL_ROM

O A[4:0]

q_i__20

RTL_MUX

I0 S=1'b1
I1 S=default

O S

q_i__21

RTL_ROM

O A[4:0]

q_i__22

RTL_MUX

I0 S=1'b1
I1 S=default

O S
q_i__23

RTL_ROM

O A[4:0]

q_i__24

RTL_MUX

I0 S=1'b1
I1 S=default

O S

q_i__25

RTL_MUX

I0[12:0] S=16'b1100001101001111

I1[12:0] S=default O[12:0]

S[15:0]

q_i__26

RTL_ROM

O A[3:0]

q_i__27

RTL_MUX

I0 S=1'b1
I1 S=default

O S

q_i__28

RTL_ROM

O A[3:0]

q_i__29

RTL_MUX

I0 S=1'b1
I1 S=default

O S

q_i__30

RTL_ROM

O A[3:0]

q_i__31

RTL_MUX

I0 S=1'b1
I1 S=default

O S

q_i__32

RTL_ROM

O A[3:0]

q_i__33

RTL_MUX

I0 S=1'b1
I1 S=default

O S

q_i__34

RTL_ROM

O A[3:0]

q_i__35

RTL_MUX

I0 S=1'b1
I1 S=default

O S

q_i__36

RTL_ROM

O A[3:0]

q_i__37

RTL_MUX

I0 S=1'b1
I1 S=default

O S

q_i__38

RTL_ROM

O A[3:0]

q_i__39

RTL_MUX

I0 S=1'b1
I1 S=default

O S

q_i__40

RTL_ROM

O A[3:0]

q_i__41

RTL_MUX

I0 S=1'b1
I1 S=default

O S

q_i__42

RTL_ROM

O A[3:0]

q_i__43

RTL_MUX

I0 S=1'b1
I1 S=default

O S

q_i__44

RTL_ROM

O A[3:0]

q_i__45

RTL_MUX

I0 S=1'b1
I1 S=default

O S

q_i__46

RTL_ROM

O A[3:0]

q_i__47

RTL_MUX

I0 S=1'b1
I1 S=default

O S

q_i__48

RTL_ROM

O A[3:0]

q_i__49

RTL_MUX

I0 S=1'b1
I1 S=default

O S

q_i__50

RTL_ROM

O A[3:0]

q_i__51

RTL_MUX

I0 S=1'b1
I1 S=default

O S

q_i__52

RTL_MUX

I0[12:0] S=16'b1100001101001111

I1[12:0] S=default O[12:0]

S[15:0]

q_reg[12:0]

RTL_REG_SYNC C

CED Q

RST

SET

s1_reg[12:0]

RTL_REG CD Q

s2_reg[12:0]

RTL_REG CD Q
tick_n_i

RTL_MUX

I0[15:0] S=16'b1100001101001111

I1[15:0] S=default O[15:0]

S[15:0]

tick_n_reg[15:0]

RTL_REG CD Q

tick_n0_i +RTL_ADD

I1

I0[15:0] O[15:0]